

9/4/2026

The law of quadratic reciprocity

Thm (Quadratic reciprocity) Let p, q be distinct primes. Then

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}}.$$

Equivalently,

$$\left(\frac{q}{p}\right) = \left(\frac{p}{q}\right),$$

unless $p, q \equiv 3 \pmod{4}$, in which case

$$\left(\frac{q}{p}\right) = -\left(\frac{p}{q}\right).$$

Eg. (1) $p = 3$, $q = \overset{1 \pmod{4}}{13}$: $\left(\frac{3}{13}\right) = \left(\frac{13}{3}\right) = \left(\frac{1}{3}\right) = 1$.

(2) $p = 3$, $q = \overset{3 \pmod{4}}{7}$: $\left(\frac{3}{7}\right) = -\left(\frac{7}{3}\right) = -\left(\frac{1}{3}\right) = -1$.

Lattice point counting.

What is an integer lattice in $\mathbb{R}^2$?

Algebraically, it is a set of the form

$$L = \{x\vec{v}_1 + y\vec{v}_2 : x, y \in \mathbb{Z}\},$$

where $\vec{v}_1, \vec{v}_2 \in \mathbb{R}^2$ are linearly independent vectors, so

that

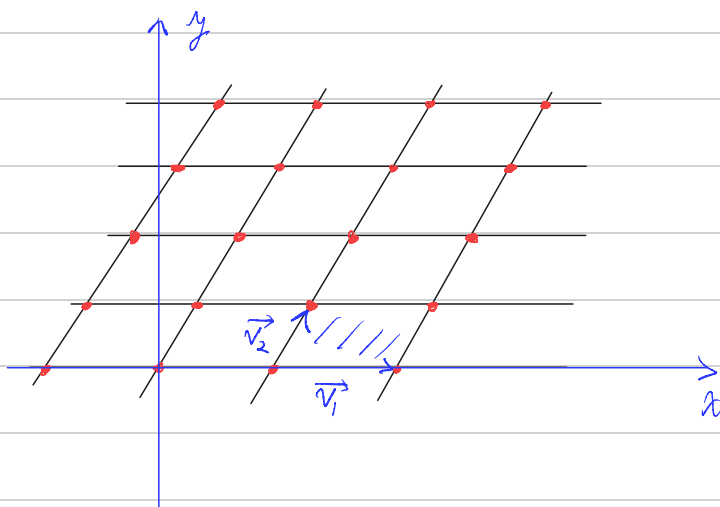
$$\mathbb{R}^2 = \{x\vec{v}_1 + y\vec{v}_2 : x, y \in \mathbb{R}\}.$$



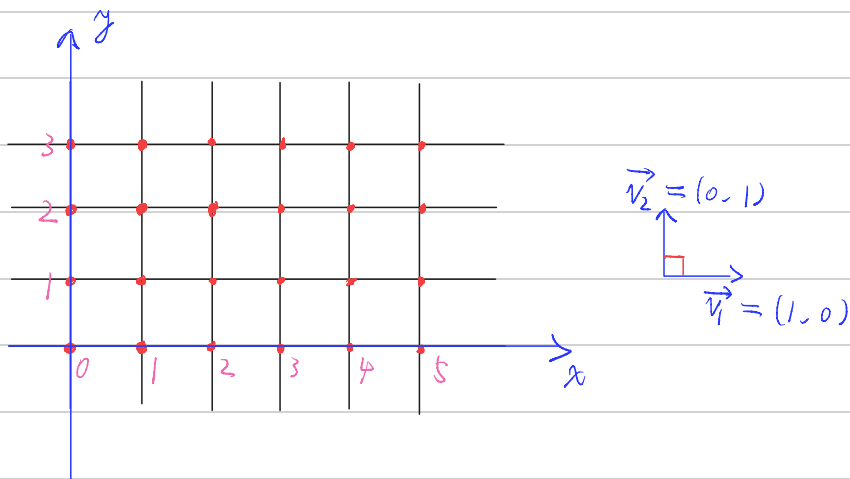
Geometrically, it consists of points on a grid formed by

infinitely many identical parallelograms.



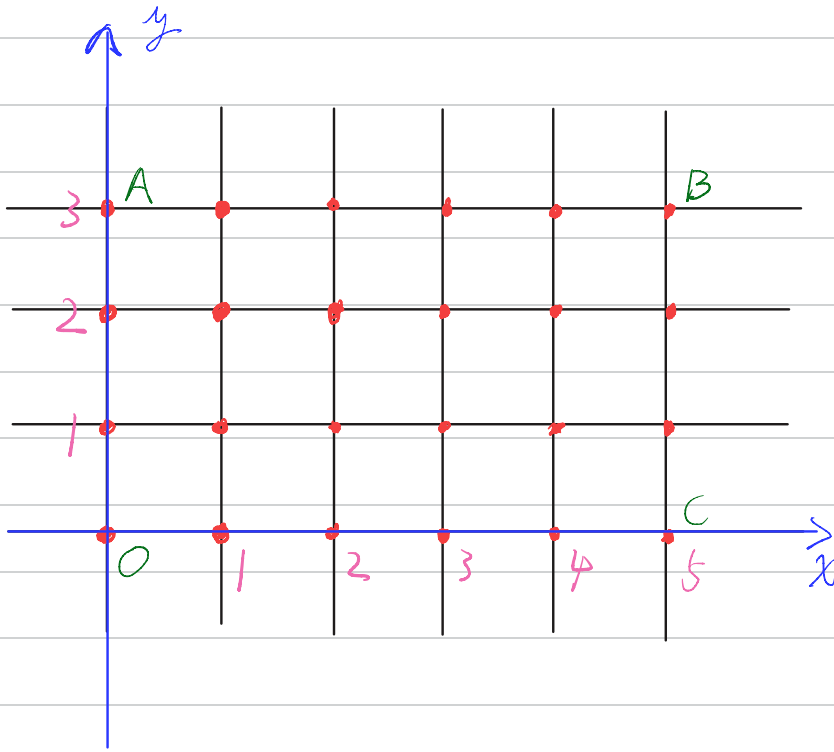


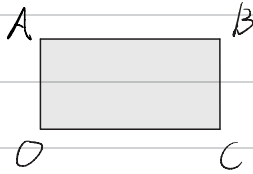
We will work with the square grid!



How to count lattice points?

1. Lattice points in a rectangle



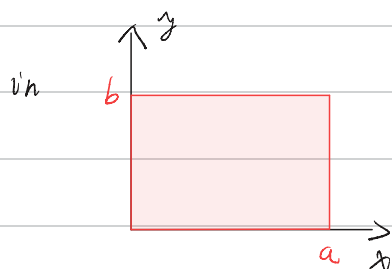
Q1: How many lattice points in  , including

points on AB, BC but not those on OC, OA?

$$A: 15 = 5 \cdot 3 = \text{Area} \left(\begin{array}{c} A \\ \square \\ O \quad C \end{array} B \right) : \quad \bullet \iff \square \rightarrow \text{Area 1}$$

Let a, b be positive integers.

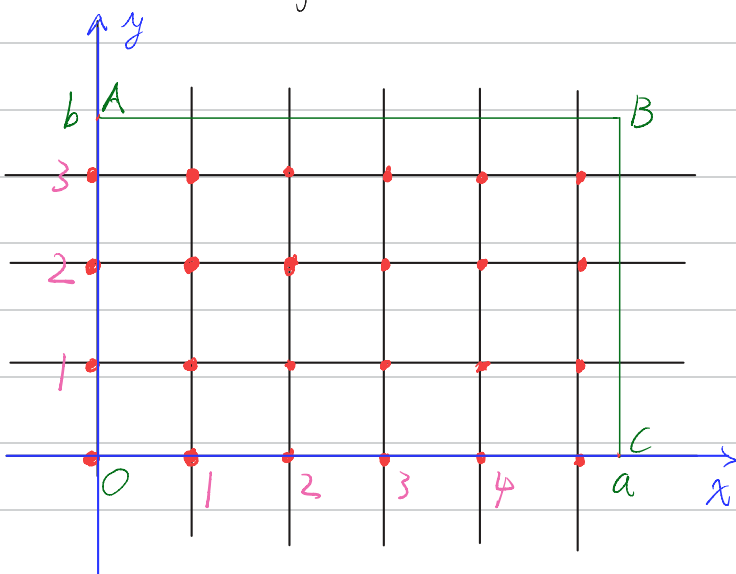
Number of lattice points



$$= \text{Area} = ab.$$

excluding those on the axes

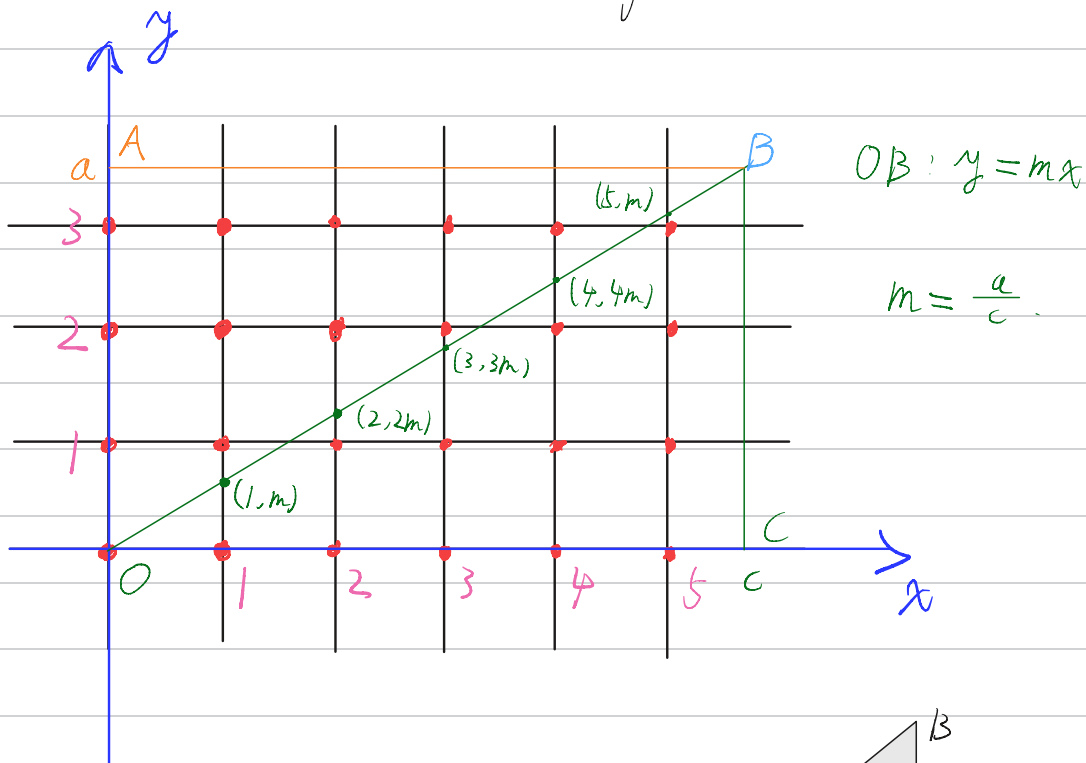
Q2: What if $a, b > 0$ are real numbers?



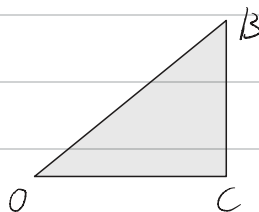
Number of lattice points

$$= \lfloor a \rfloor \cdot \lfloor b \rfloor.$$

2. Lattice points in a triangle



Q1: How many lattice points in

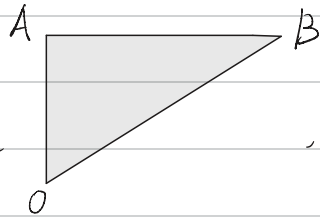


, including

points on OB, BC but not those on OC?

$$A: \lfloor 1 \cdot m \rfloor + \lfloor 2 \cdot m \rfloor + \lfloor 3 \cdot m \rfloor + \dots + \lfloor \lfloor c \rfloor m \rfloor = \sum_{k=1}^{\lfloor c \rfloor} \left\lfloor \frac{ka}{c} \right\rfloor$$

Q2: How many lattice points in , including



points on OB, AB but not those on OA?

$$A: \lfloor 1 \cdot \frac{1}{m} \rfloor + \lfloor 2 \cdot \frac{1}{m} \rfloor + \dots + \lfloor |a| \cdot \frac{1}{m} \rfloor = \sum_{k=1}^{|a|} \lfloor k \cdot \frac{1}{m} \rfloor = \sum_{k=1}^{|a|} \lfloor \frac{kc}{a} \rfloor$$

using OB: $x = \frac{1}{m} \cdot y$.

Recall :

Gauss's lemma: Let $p > 2$ be prime and let $a \in \mathbb{Z}$ with $p \nmid a$. Define

$$\varepsilon_a = \begin{cases} 1 & , \quad a > 0, \\ -1 & , \quad a < 0. \end{cases}$$

Then

$$\left(\frac{a}{p}\right) = (-1)^{S(a,p)} \varepsilon_a^{\frac{p-1}{2}},$$

where

$$S(a,p) = \sum_{k=1}^{\frac{p-1}{2}} \left\lfloor \frac{k \varepsilon_a a}{p} \right\rfloor + \frac{p^2-1}{8} (a-1).$$

Note: If a is odd, then $\frac{p^2-1}{8} (a-1) \equiv 0 \pmod{2}$.

Eisenstein's proof of quadratic reciprocity:

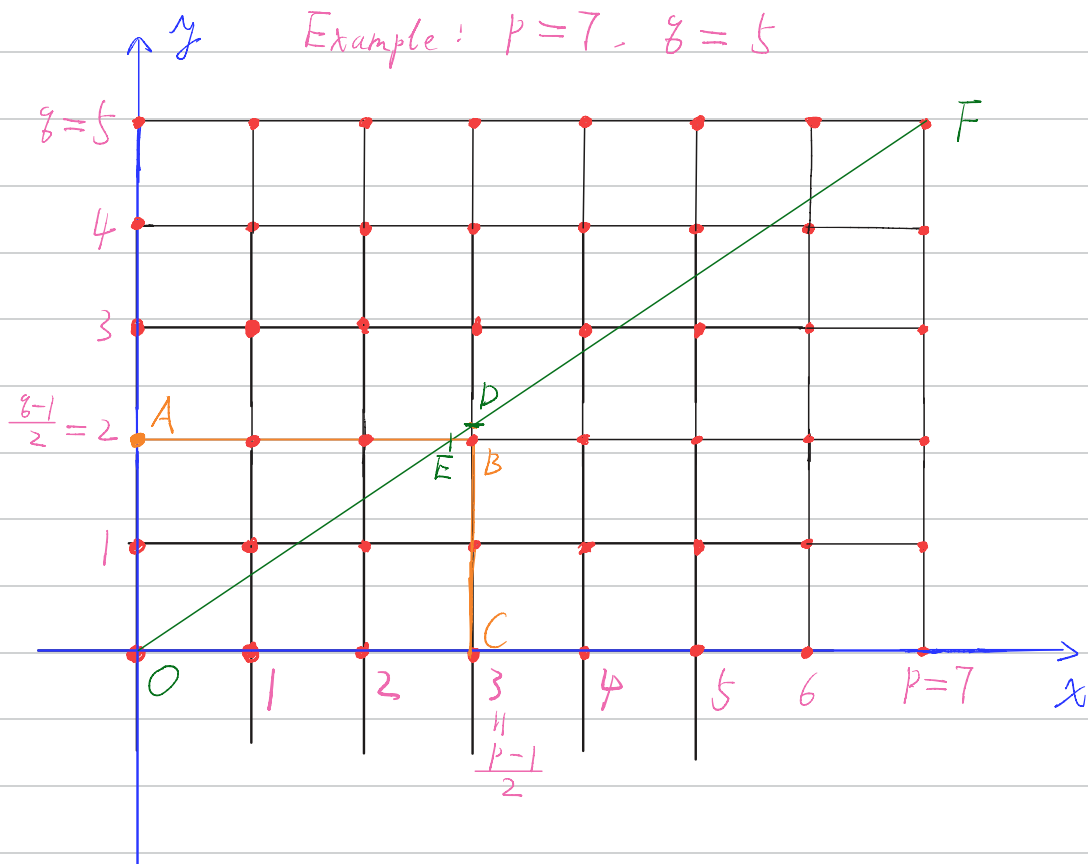
By Gauss's lemma,

$$\left(\frac{p}{q}\right) = (-1)^{S(p,q)}, \quad S(p,q) = \sum_{k=1}^{\frac{q-1}{2}} \left\lfloor \frac{kp}{q} \right\rfloor.$$

$$\left(\frac{q}{p}\right) = (-1)^{S(q,p)}, \quad S(q,p) = \sum_{k=1}^{\frac{p-1}{2}} \left\lfloor \frac{kq}{p} \right\rfloor.$$

$$\Rightarrow \left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{S(p,q) + S(q,p)}$$

$$\stackrel{?}{=} (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}}.$$



① Number of lattice points in $\begin{matrix} A & B \\ | & | \\ O & C \end{matrix} = \frac{p-1}{2} \cdot \frac{g-1}{2}$